Introduction to (Statistical) Machine Learning
Problem Set One

Summer 2026, IIM B

Due by 11:59 pm on 29 June 2026

Instructions. Problems 1–4 are required and should be completed by all students. Problems 5–7 are optional and are intended for students who would like additional practice or wish to explore some of the ideas from class in greater depth. Students are encouraged to attempt the optional problems. For theoretical problems, using LLM's is not allowed. You are free to use them for the coding problems.

1. Consider a clinical test for cancer that can yield either a positive (+) or negative (-) result. Suppose that a patient who truly has cancer has a 1% chance of slipping past the test undetected. On the other hand, suppose that a cancer-free patient has a 5% probabiliity of getting a positive test result. Suppose also that 2% of the population has cancer. Suppose that a patient who has taken the test **twice** has tested positive both times.

   a) What is the probability that this patient has cancer?

   b) What is the probability that this patient will test positive if they take the test the third time?

   Describe clearly the probability model you are using to calculate the above probabilities.

2. We have an urn with N balls of which θ are red and the remaining $N - \theta$ are white. We are assuming that N is known but θ is unknown. We sample n balls from the urn without replacement and suppose we see x red balls in the sample. What can we infer from θ based on knowledge of x, n and N?

   This problem has applications in survey sampling. Suppose we want to know the number θ in a city (of total population N) that support a particular political party. We take a sample of n people out of which x support that party. What can we then say about θ?

   a) Argue that the likelihood function is given by:

   $$\mathbb{P}\{X = x \mid N, \theta, n\} = \frac{\binom{\theta}{x}\binom{N-\theta}{n-x}}{\binom{N}{n}}$$

   This is the Hypergeometric distribution (`https://en.wikipedia.org/wiki/Hypergeometric_distribution`).

b) Work with the prior:

$$\mathbb{P}\{\theta = \theta \mid N\} = \frac{1}{N+1} I\{\theta \in \{0, 1, \dots, N\}\},$$

and show that the posterior is given by

$$\mathbb{P}\{\theta = \theta \mid x, N, n\} = \frac{\binom{\theta}{x}\binom{N-\theta}{n-x}}{\binom{N+1}{n+1}} I\{\theta \in \{0, 1, \dots, N\}\}.$$

c) Show that

$$\mathbb{E}(\theta \mid x, n, N) = \frac{(N+2)(x+1)}{n+2} - 1$$

and

$$\operatorname{var}(\theta \mid x, n, N) = \frac{p(1-p)}{n+3}(N+2)(N-n) \qquad \text{where } p := \frac{x+1}{n+2}.$$

d) When N is large, argue that

$$\mathbb{E}\left(\frac{\theta}{N} \mid x, n, N\right) \approx \frac{x+1}{n+2} = p$$

and

$$\operatorname{var}\left(\frac{\theta}{N} \mid x, n, N\right) \approx \frac{p(1-p)}{n+3}.$$

This implies that, when N is large, a point estimate of θ/N can be taken to be p and the uncertainty of this point estimate can be taken to be the posterior standard deviation $\sqrt{\frac{p(1-p)}{n+3}}$.

e) Let R_{n+1} denote the event that the next draw (i.e., the $(n+1)$-th draw) leads to a red ball. Show that the posterior probability of R_{n+1} given x, n, N is given by:

$$\mathbb{P}\{R_{n+1} \mid x, n, N\} = \frac{x+1}{n+2} = p.$$

This formula is known as the Laplace Rule of Succession.

3. We discussed the problem of estimating θ from numerical measurements $x_1, \dots, x_n$. We used the following Bayesian model:

$$X_1, \dots, X_n \mid \theta, \sigma \overset{\text{i.i.d}}{\sim} N(\theta, \sigma^2) \quad \text{and} \quad \theta, \log\sigma \overset{\text{i.i.d}}{\sim} \text{uniform}(-\infty, \infty).$$

We proved that the posterior density of θ under this model satisfies:

$$\frac{\sqrt{n}\,(\theta - \bar{x})}{\hat{\sigma}} \mid \text{data} \sim t_{n-1} \tag{1}$$

where

$$\bar{x} := \frac{x_1 + \cdots + x_n}{n} \quad \text{and} \quad \hat{\sigma}^2 := \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

In this exercise, we shall go over a slightly different proof of the same result. Along the way, we shall also discuss inference for σ.

a) Show that

$$\theta \mid \text{data}, \sigma \sim N(\bar{x}, \frac{\sigma^2}{n}).$$

The left hand side above refers to the conditional distribution of θ given the data as well as the parameter σ.

b) Show that the posterior density of σ satisfies:

$$f_{\sigma|\text{data}}(\sigma) \propto \sigma^{-n} \exp\left(-\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{2\sigma^2}\right) I\{\sigma > 0\}.$$

c) Use the results of the previous two parts along with the formula:

$$f_{\theta|\text{data}}(\theta) = \int_0^\infty f_{\theta|\text{data},\sigma}(\theta) f_{\sigma|\text{data}}(\sigma) d\sigma$$

to deduce (1).

d) Use the result in part (b) to show that

$$\frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \bar{x})^2 \mid \text{data} \sim \chi^2_{n-1}. \tag{2}$$

On this basis, comment on whether $\hat{\sigma} = \sqrt{\sum_{i=1}^n (x_i - \bar{x})^2/(n-1)}$ is a reasonable point estimate of σ

e) Using (2), derive a $100(1-\alpha)\%$ Bayesian credible interval for σ for fixed α.

f) Suppose $n = 6$ and $x_1 = 26.6, x_2 = 38.5, x_3 = 34.4, x_4 = 34, x_5 = 31, x_6 = 23.6$. Explicitly calculate the 95% credible interval for σ.

4. We considered the frequentist probability:

$$\mathbb{P}\left\{\bar{X}_N - \frac{t_{N-1,\alpha/2}}{\sqrt{N}}\sqrt{\frac{1}{N-1}\sum_{i=1}^N (X_i - \bar{X})^2} \leq \theta \leq \bar{X}_N + \frac{t_{N-1,\alpha/2}}{\sqrt{N}}\sqrt{\frac{1}{N-1}\sum_{i=1}^N (X_i - \bar{X})^2}\right\} \tag{3}$$

where $X_1, X_2, \dots$ are i.i.d $N(\theta, \sigma^2)$ (for fixed θ and σ), and N is the stopping rule:

$$N := \inf\{n \geq 1 : X_n \leq 25\}. \tag{4}$$

It was mentioned in class that, unlike the case when N is a fixed number such as 6, the probability (3) will be different from $1 - \alpha$. The goal of this problem is to verify this by simulation.

The probability (3) is actually not well-defined because of the presence of $N - 1$ in the denominator (as well as $N - 1$ degrees of freedom for the t-quantile), as N in (4) can equal 1 with some positive probability. We therefore attempt to calculate the simpler probability:

$$\mathbb{P}\left\{\bar{X}_N - \sigma\frac{z_{\alpha/2}}{\sqrt{N}} \leq \theta \leq \bar{X}_N + \sigma\frac{z_{\alpha/2}}{\sqrt{N}}\right\}. \tag{5}$$

Follow the steps below to write code to calculate the above probability.

a) Fix $\theta = 35.5$ and $\sigma = 5.5$.

b) Set $\alpha = 0.05$ and $M = 100000$.

c) For $i = 1, \ldots, M$:

- Generate $N(\theta, \sigma^2)$ random variables $X_1, X_2, \ldots$ until an observation less than 25 is obtained. Let N be the number of observations.
- Calculate $C_i = I\left\{\bar{X}_N - \sigma\frac{t_{N-1,\alpha/2}}{\sqrt{N}} \le \theta \le \bar{X}_N + \sigma\frac{t_{N-1,\alpha/2}}{\sqrt{N}}\right\}$.

d) The estimate for the probability (5) is $(C_1 + \cdots + C_M)/M$.

Write code and compute this estimate of (5). Comment on how different the estimate is from $1 - \alpha$.

5. Consider again the problem of estimating an unknown physical quantity θ from data $x_1 = 26.6, x_2 = 38.5, x_3 = 34.4, x_4 = 34, x_5 = 31, x_6 = 23.6$. Suppose that the measurement device can only record observations if they are in the range $[23, 39]$. Based on this information and the data, what can be inferred about θ?

   a) Argue that the likelihood now should be:
   $$\prod_{i=1}^{n} \frac{\frac{1}{\sigma}\phi\left(\frac{x_i-\theta}{\sigma}\right)}{\Phi\left(\frac{39-\theta}{\sigma}\right) - \Phi\left(\frac{23-\theta}{\sigma}\right)}$$
   where $\phi(z) := (2\pi)^{-1/2}\exp(-z^2/2)$ is the normal density and $\Phi(z) = \int_{-\infty}^{z}\phi(u)du$ is the normal cdf.

   b) Using the same prior $\theta, \log\sigma \overset{\text{i.i.d}}{\sim} \text{uniform}(-\infty, \infty)$, write down the joint posterior density for θ and σ.

   c) Construct a dense grid of points $(\theta, \log\sigma)$ (say in the range $[15, 50] \times [-1, 3]$). On this grid, compute the unnormalized posterior density, and normalize numerically.

   d) Based on the discretized posterior, compute an approximate 95% Bayesian credible interval for θ. Compare your interval to the interval obtained by the usual normal analysis from class which does not take the truncation into consideration. For example, does this analysis with the truncated prior lead to a smaller credible region?

6. Consider again the problem of estimating an unknown physical quantity θ from data. Suppose the measurement device works correctly if the measurement lies in the range $[23, 39]$. If the measurement falls outside this range, then the device displays `Failure`. Suppose we got the observations: $x_1 = 26.6, x_2 = 38.5, x_3 = 34.4, x_4 = 34, x_5 = 31, x_6 = 23.6, x_7 =$ `Failure`, $x_8 =$ `Failure`. Based on this information and the data, what can be inferred about θ?

   a) Argue that the likelihood now should be:
   $$\left[\prod_{i=1}^{6}\frac{1}{\sigma}\phi\left(\frac{x_i-\theta}{\sigma}\right)\right]\left[1 - \Phi\left(\frac{39-\theta}{\sigma}\right) + \Phi\left(\frac{23-\theta}{\sigma}\right)\right]^2$$
   where $\phi(z) := (2\pi)^{-1/2}\exp(-z^2/2)$ is the normal density and $\Phi(z) = \int_{-\infty}^{z}\phi(u)du$ is the normal cdf.

   b) Using the same prior $\theta, \log\sigma \overset{\text{i.i.d}}{\sim} \text{uniform}(-\infty, \infty)$, write down the joint posterior density for θ and σ.

c) Construct a dense grid of points $(\theta, \log \sigma)$ (say in the range $[15, 50] \times [-1, 3]$). On this grid, compute the unnormalized posterior density, and normalize numerically.

d) Based on the discretized posterior, compute an approximate 95% Bayesian credible interval for θ. Compare your interval to the interval obtained by the usual normal analysis from class which does not take the truncation into consideration. For example, does this analysis with the truncated prior lead to a smaller credible region?

7. Consider yet again the problem of estimating an unknown physical quantity θ from observations $x_1, \dots, x_n$. Now consider the following Bayesian model:

$$X_1, \dots, X_n \mid \theta, \sigma \overset{\text{i.i.d}}{\sim} \text{Lap}(\theta, \sigma) \quad \text{and} \quad \theta, \log \sigma \overset{\text{i.i.d}}{\sim} \text{uniform}(-\infty, \infty).$$

Here $\text{Lap}(\theta, \sigma)$ denotes the Laplace distribution which has density

$$x \mapsto \frac{1}{2\sigma} \exp\left(-\frac{|x - \theta|}{\sigma}\right).$$

a) Show that the posterior of θ satisfies:

$$f_{\theta|\text{data}}(\theta) \propto \left(\frac{1}{M(\theta)}\right)^n \qquad \text{where } M(\theta) := \sum_{i=1}^{n} |x_i - \theta|.$$

b) For the data $x_1 = 26.6, x_2 = 38.5, x_3 = 34.4, x_4 = 34, x_5 = 31, x_6 = 23.6$, evaluate the density at a dense grid of points in some large range (e.g., $[15, 50]$), normalize correctly so that the integral of the posterior density is 1, and then plot the posterior density. Comment on the nature of the plot.

c) Use your discretized approximation of the posterior to construct an approximate 95% credible interval for θ. Compare your interval with the interval derived from the normal based method from class.

d) Repeat the exercise from part (c) when the data is $x_1 = 26.6, x_2 = 38.5, x_3 = 34.4, x_4 = 34, x_5 = 31, x_6 = 23.6, x_7 = 120$. The additional data point $x_7 = 120$ is an outlier, and the goal here is to see how the outlier affects the two intervals based on the normal and Laplace models.